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Research on the application of data mining technology in software engineering

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Abstract

The role of data mining in software engineering is obvious, but there is a lack of mining depth. In the past, engineering layout and function distribution problems in software development, and the framework construction was unreasonable. Therefore, this paper proposes a software system based on data mining technology. First, the software engineering project is divided, and the software engineering project design is carried out according to the interactive requirements to realize the preprocessing of software engineering. Then, according to the design criteria, the design collection of software engineering is formed, and the information system is analyzed in depth. MATLAB simulation shows that the design interactivity and accuracy of data mining are superior to traditional design methods under the condition that software engineering requirements are consistent.

Keywords: *data mining; interactivity; Framework; Software Engineering*

1.Introduction

Software engineering design is one of the important projects of software design and plays a very important role in the improvement of the system[1]. However, in the process of software system design[2], the design system has the problem of digging depth, and cannot effectively play the system function[3]. Some scholars believe that the application of data mining to software engineering design can effectively carry out functional perfection and interactive analysis[4], and provide corresponding support for software system verification[5]. On this basis, this paper proposes a software engineering design method based on data mining technology and verifies the effectiveness of the holiday.

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2. Related concepts

Software development is to use software key points, software solutions, the relationship between software functions, and system standards to design software engineering[6], and build functions according to design indicators, and form a design table.

2.1 Mathematical description of software development

Through the integration of design functions, the perfection of the design system is finally judged[7]. The application of data mining combined with system standards to design development system projects can improve the level of software design[8].

Hypothesis 1: Software engineering is x_i, the design result is $raeal(x_i)$, the design criterion is $de(x_i)$, and the software design function is y_i,which as shown in Equation (1).

$$de(x_i) = \sum x \to y$$

2.2 Selection of Software Features

Hypothesis 2: The functional interaction function is $jc(x_i)$ [9]and the software design adjustment factor is κ [10], then the software engineering design is shown in Equation (2).

$$jc(x_i)=de(x_i) \otimes \kappa \quad (2)$$

2.3 Processing of unreasonable data in software engineering

Before data mining and analysis, it is necessary to conduct standard analysis of the functions in the design system to obtain a more reasonable design scheme[11]. First, the overall requirements of software engineering are comprehensively analyzed[12], and the interaction logic of software functions is set to provide support for the application of data mining. The software design needs to be preprocessed[13], and if the processed results meet the requirements of software engineering[14], the processing is effective, otherwise the functional structure is re-deepened[15]. In order to improve the accuracy of function mining, improve the interactivity and compatibility of functions, the function mining scheme should be selected[16], and the specific method selection is shown in Figure 1.

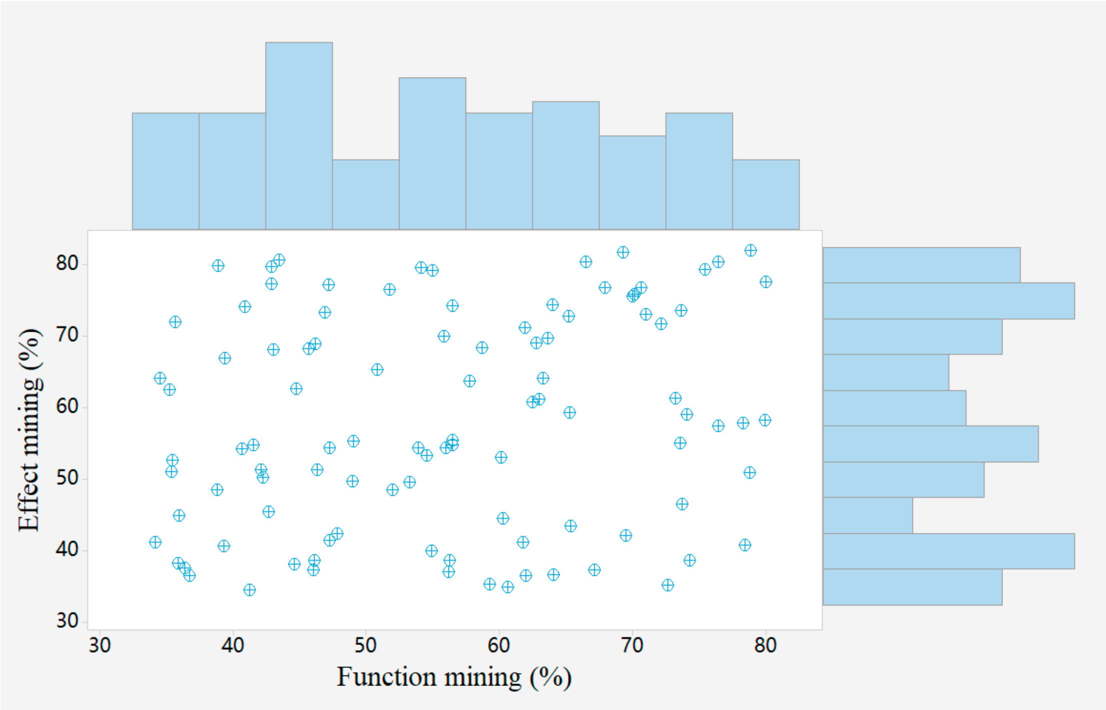

Fig.1.Functional mining analysis results

The software engineering in Figure 1 shows that the functional development results are homogeneous and meet the objective system requirements[17]. The functional perfection is not directional, indicating that the functional mining analysis has high accuracy, so it is studied as software engineering[18]. The functional perfection meets the development requirements, mainly because the system standard adjusts the functional perfection, removes the repeated unreasonable software engineering[19], and revises the key points of the software engineering, so that the selectivity of the entire software engineering is high.

3. The perfection of different software functions

Function mining rationally analyzes software functions[20], and adjusts the corresponding unreasonable software engineering relationships to realize the dynamic interaction design of software engineering[21]. Function mining divides software designs into different categories and randomly selects different design schemes for verification[22]. In the process of software engineering, the interaction requirements of the transfer class are related to the functional perfection[23]. After the correlation processing is completed, compare different methods for software engineering, and the most interactive software features.

4. Practical examples of software design

4.1 Software design

In order to facilitate software design analysis, different software designs are studied in this paper, as shown in Table 1.

Tab.1.Parameters of software engineering

Software design	Response time	Implement functionality	Design standards
compatibility	part	60.33	74.49

	Big picture	58.33	69.80
Processing power	part	44.27	58.70
	Big picture	57.99	49.77
Calculate the speed	part	54.59	75.40
	Big picture	69.33	46.20

The processing of the software engineering planning function in Table 1 is shown in Table 2.

Tab.2.Processing results of software engineering planning functions

outcome	Local features	Overall functionality	compatibility	Data heterogeneity
Data mining techniques	56.45	74.44	69.75	68.52
Theoretical framework design	62.74	57.18	54.67	60.10

Table 1 shows that compared with traditional software design, the design system of application function mining is closer to the actual interactivity[24]. In terms of interactivity and freedom of software design, functional mining is better.

4.2 Interactivity of software design

The design of software engineering includes the space occupancy, system complexity, etc., and the design of different interaction scales. After the preprocessing of application function mining, preliminary software engineering is obtained, and the perfection of the design results is analyzed[25]. In order to verify the effect more accurately, different unreasonable software projects are selected to calculate the overall interactivity of the software design, as shown in Table 3.

Tab.3.Optimized interactivity of software design

function	Local optimization	Overall optimization
correspondence	77.30	55.91
personality	63.27	71.29
transmit	42.62	60.19
mean	56.51	56.26
X2	66.53	58.35
P=3.022		

4.3 Accuracy of Software Design

In order to verify the accuracy of the functional design, the accuracy design and accuracy comparison are made with the software design. It can be seen from Table 4 that the accuracy design of functional design is shorter than that of traditional design methods, but the error rate is lower, indicating that the choice of functional design is relatively stable[26], while the design accuracy of traditional design methods is uneven. The accuracy of the above algorithm is shown in Table 4.

Tab.4.Comparison of design accuracy of different methods

algorithm	Design accuracy	Design accuracy	error
Data mining techniques	51.17	47.35	0.68
	52.25	49.39	0.59
	46.61	55.83	0.64
	59.71	59.65	0.72
	45.18	45.98	0.45
	56.19	47.72	0.66
	57.69	57.27	0.72
	48.17	57.09	0.20
	52.72	54.26	0.32
	55.54	59.50	0.79

	46.70	50.29	0.91
	49.68	58.05	0.68
Traditional design methods	55.98	57.22	0.59
	58.32	47.02	0.64
	47.64	53.48	0.72
	53.06	55.20	0.45
	50.17	47.83	0.66
	54.63	52.00	0.72
	57.26	49.60	0.20
	55.01	59.72	0.32
	52.86	58.07	0.79
	51.17	47.35	0.91
	52.25	49.39	0.68
	46.61	55.83	0.59
	44.44	42.44	44.44
P	0.041	0.023	0.026

Table 3 shows that the data mining technology has deficiencies in accuracy design and accuracy at the level of functional interaction, and the accuracy of the function implementation changes greatly and the delay rate is low. The accuracy of the dynamic interaction results of functional design is higher than that of software design. At the same time, the accuracy design of the functional design is greater than 60%, and the accuracy has not changed significantly. To further verify the superiority of the functional design. In order to further verify the design effect of the method on software engineering, the dynamic interaction analysis of software design is carried out by different methods, and the results are shown in Figure 2.

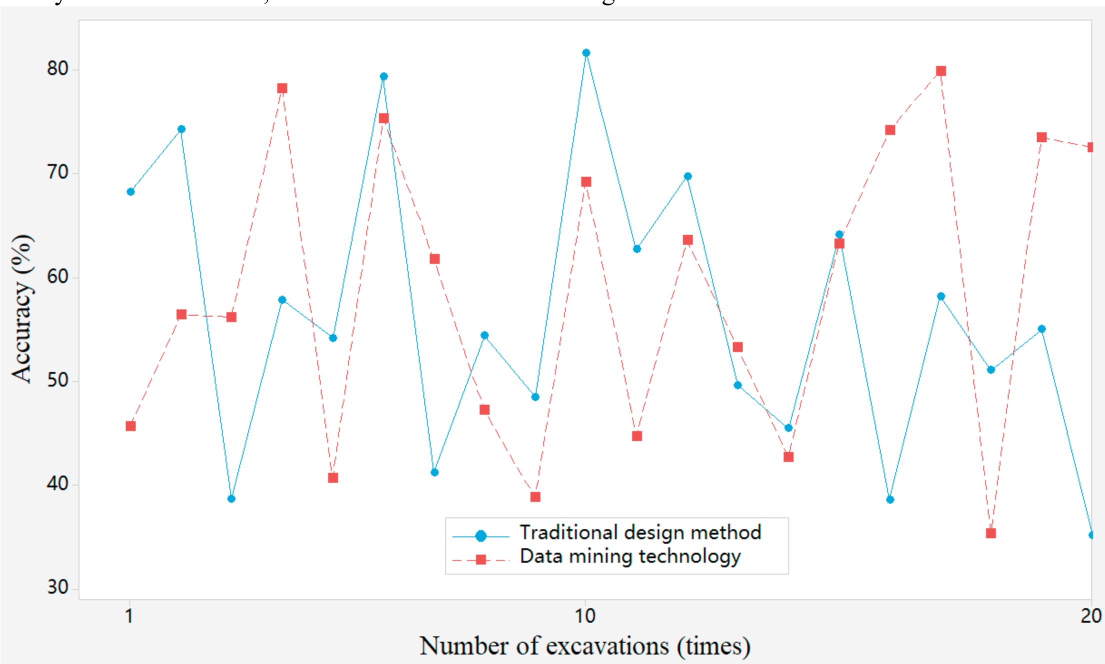

Fig.2.Dynamic accuracy of software design

It can be seen from Figure 2 that data mining technology is significantly better than traditional design methods, and the reason is that the functional design increases the accuracy adjustment coefficient, and sets the corresponding functional interaction logic to remove the results that do not meet the requirements.

4.3 Interactivity of software design

To verify the interactivity of the functional design, compare it with previous design methods. It can be seen from Table 5 that the interactive design of functional design is shorter than that of traditional design methods, but the error rate is lower, indicating that the choice of functional design is relatively stable, while the design interactivity of traditional design methods is uneven. The interactivity of the above algorithm is shown in Table 5.

Tab.5.Design interactivity comparison of different methods

algorithm	Random test	Design interactivity	Design interactivity	Interactive redundancy
Data mining techniques	2	42.68	44.74	0.77
	5	34.11	47.75	0.92
	53	70.26	46.47	0.34
	34	73.74	53.17	0.12
	38	78.56	43.31	0.65
	34	48.87	72.23	0.31
	29	79.14	69.34	0.75
	14	53.01	51.00	0.45
	34	65.82	48.58	0.86
	19	64.36	38.26	0.17
	28	69.54	60.64	0.10
	55	70.83	75.80	0.93
	6	75.92	36.25	0.93
	45	59.09	65.66	0.48
	55	41.53	61.77	1.00
Traditional design methods	12	60.09	68.39	0.35
	54	47.22	53.07	0.62
	33	37.26	40.44	0.22
	40	76.55	78.98	0.97
	45	74.59	61.32	0.16
	2	48.95	69.61	0.93
	52	44.17	58.58	0.12
	9	42.68	44.74	0.77
	40	34.11	47.75	0.92
	12	70.26	46.47	0.34
	29	73.74	53.17	0.12
	42	78.56	43.31	0.65
	39	48.87	72.23	0.31
	10	79.14	69.34	0.75
	51	53.01	51.00	0.45
	16	65.82	48.58	0.86
P		8.141	9.123	8.256

It can be seen from Table 3 that data mining technology has deficiencies in interactive design and interactivity at the level of functional interaction, and the interactivity of functional implementation has changed significantly, and the interaction is redundant. The dynamic interaction results of functional design are higher than those of software design. At the same time, the interactive design of functional design is greater than 60%, and the accuracy has not changed significantly. To further verify the superiority of the functional design. Dynamic interaction analysis of software design using different methods is shown in Figure 3.

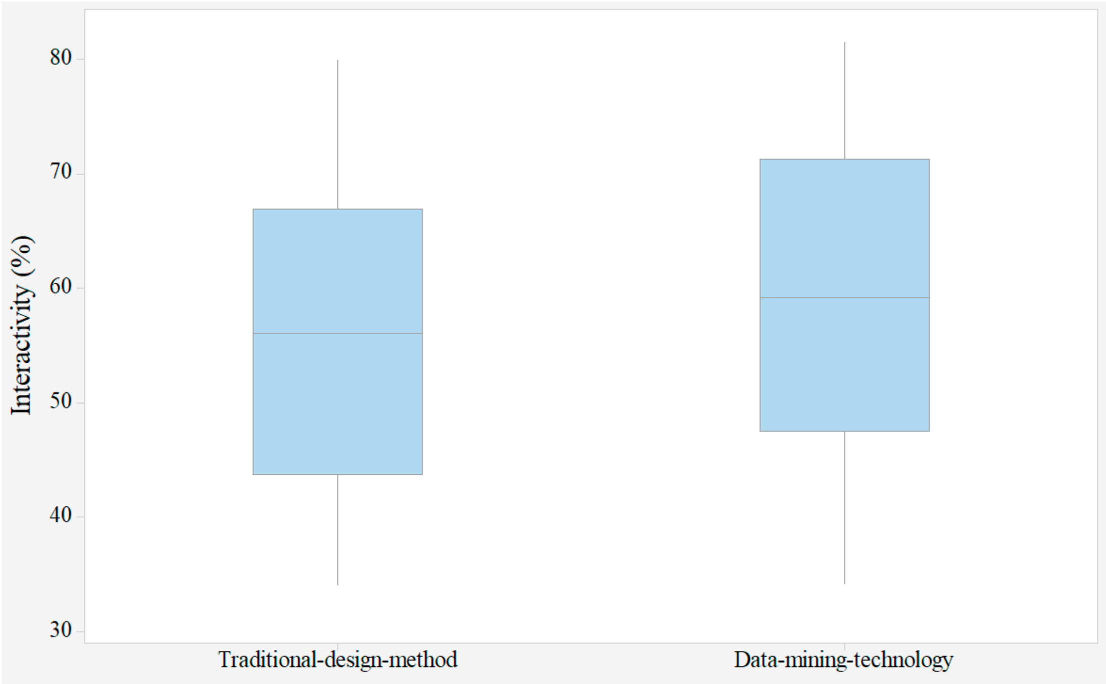

Fig.3.Dynamic interaction results of software design

It can be seen from Figure 3 that data mining technology is significantly better than traditional design methods, and the reason is that the functional design adds an interactive buffer module and sets the corresponding interaction logic to simplify the interaction process.

5. Conclusion

In the case of increasing requirements of software engineering, in view of software design problems, this paper proposes functional design and combines software engineering standards to improve software functions. At the same time, the design system is analysed in depth to form a design collection. Studies have shown that functional design can improve the interactivity and interactivity of software design and realize the dynamic interaction design of software engineering. However, in the process of application function design, too much attention is paid to design indicators and dynamic interaction index analysis is ignored.

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